Laboratory for Electrical Instrumentation and Embedded Systems
IMTEK – Department of Microsystems Engineering
University of Freiburg

Sensors Lab Course
Winter term 2025/26

Lab Report M1

Acceleration and Pressure Sensors

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November 28, 2025

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1 Introduction

Navigation tasks are typically handled using acceleration sensors, which provide data for calculating related values. In this module, the same process is explored using a high precision pressure sensor. The accelerometer and pressure sensor on the Nicla Sense ME Board are used inside an elevator to estimate the height of the building. Additional attributes of both sensors are also compared and analyzed.

2 Theory

Reliable estimation of motion and altitude can be done by observing changes in acceleration and atmospheric pressure [1]. Accelerometers use linear motion and gravitational effects, while pressure sensors measure variations in air pressure that correspond to height differences. Combining both sensor types provides a more robust description of vertical displacement, which is explored in this experiment using the sensors on the Nicla Sense ME Board [2].

2.1 Acceleration Sensor

MEMS accelerometers measure acceleration by tracking the movement of a suspended silicon mass. The proof mass is connected to the substrate using compliant suspension beams, which allow small displacements when inertial forces act on the device. Interdigitated comb electrodes are used to detect this motion. A shift of the mass changes the differential capacitance between the movable and fixed comb fingers, which is then converted into an acceleration value [3].

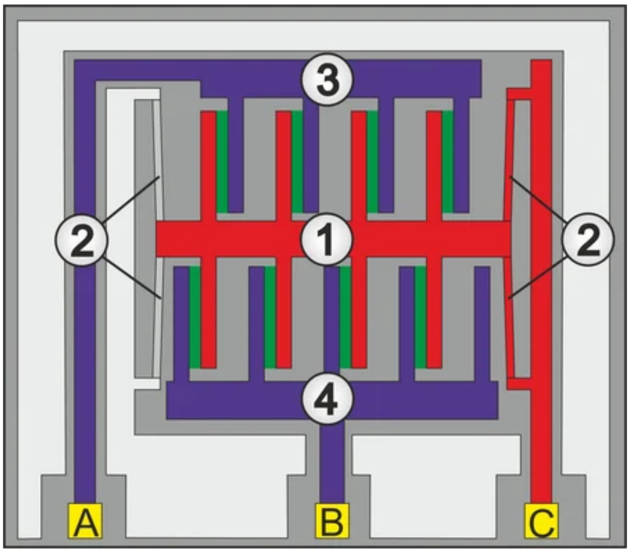

Figure 1: Mechanical model of a microfabricated capacitive accelerometer
[3]

As shown in Figure 1, the central element (1) acts as the proof mass and the beams (2) hold it in its neutral position. The fixed comb structures (3 and 4) form differential capacitors with the movable fingers. Any displacement changes the effective overlap area, resulting in a measurable capacitance variation.

To maintain stable operation, an electrostatic feedback force can be applied to keep the mass centered [1].The required force is proportional to the external acceleration.

The operating principle follows Newton's second law:

$$F = m \times a \tag{1}$$

Integrating the measured acceleration gives velocity and, with a second integration, displacement:

$$Distance = \int Velocity\, dt = \int \int Acceleration\, dt \tag{2}$$

On the Nicla Sense ME Board, acceleration is measured using the BHI260AP, which includes a 3-axis accelerometer, gyroscope, and onboard calibration and compensation.

2.2 Pressure Sensor

Atmospheric pressure changes with altitude according to the barometric relation:

$$P = P_0 \exp\left(-\frac{gMh}{RT}\right) \tag{3}$$

This formula links pressure P to reference value P_0, gravitational constant g (9.18 m/s$^{\mathbf{2}}$), height h, molar mass of air M (0.02896 kg mol^{-1}), universal gas constant R (8.314 J mol^{-1} K^{-1}) and absolute temperature T [2]

MEMS pressure sensors typically consist of a thin silicon membrane forming a diaphragm. When pressure on one side changes, the membrane deforms, and this deflection is measured using piezoresistive or capacitive sensing elements arranged in a bridge configuration. Figure 2 illustrates a common membrane based pressure sensing structure, similar to the one inside the BMP390 used in this experiment [4].

The BMP390 integrates this MEMS diaphragm with an ASIC that contains amplification, a 24 bit ADC, and temperature compensation algorithms. Its relative accuracy is approximately ± 0.03 hPa, allowing height resolution better than 0.3 m. The Nicla Sense ME exposes the compensated reading directly via its virtual barometric sensor channel [2]

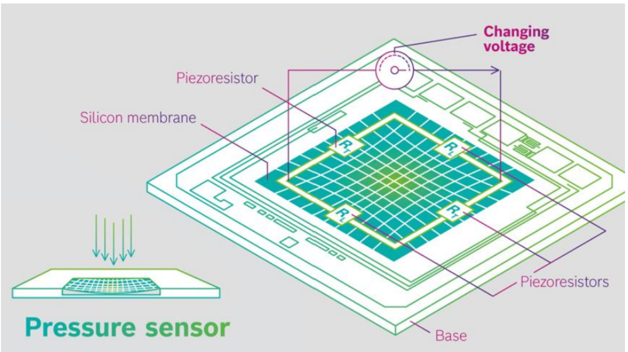

Figure 2: Pressure sensor with piezoelectric elements measuring deflection of membrane [4]

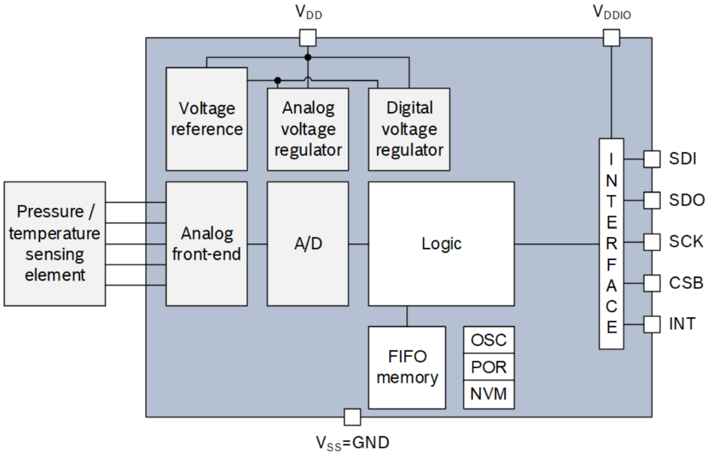

Figure 3: Block Diagram of BMP390
[2]

The metallic housing of the BMP390 contains a small opening that allows air pressure to reach the membrane. When inspected closely, the sensor can be seen to consist of two main sections: the MEMS die containing the membrane structure and the ASIC responsible for processing. These components are linked through five bond wires.

3 Methods

3.1 Setup

All acceleration measurements were obtained using the BHI260AP and all pressure measurements using the BMP390, both integrated on the Arduino Nicla Sense Board ME. The board was connected through USB to a battery powered laptop, which supplied power and acted as the serial logging interface. The firmware was developed in the Arduino IDE with the Arduino_BHY2 library version 1.5 [5]. Data was recorded using Tera Term (v. 5.5.0 x86).

Sensor outputs were accessed through the virtual sensors `SENSOR_ID_ACC_PASS` for acceleration and `SENSOR_ID_BARO` for pressure [5]. The sampling interval for all measurements was 100 ms, corresponding to 10 Hz. The accelerometer operated in its default 8g range, giving a sensitivity of $417.5\,\mathrm{LSB\,m^{-1}\,s^{2}}$ [6]. The BMP390 pressure sensor provided a relative accuracy of ± 0.03 hPa, which corresponds to a height accuracy better than ± 0.3 m [4]. Temperature during all tasks was measured using the onboard virtual sensor `SENSOR_ID_TEMP` [5].

3.2 Task 1

Pressure measurements were performed at an ambient temperature of 24°C. The sensor board was placed on a table with its Z axis pointing upward and kept stationary. Around 1000 pressure samples were recorded, each taken at a 100 ms interval.

3.3 Task 2

Acceleration data was collected under the same stationary conditions and temperature as in Task 1. A total of 1000 readings along the X, Y, and Z axes were captured with a 100 ms sampling interval.

3.4 Task 3

Measurements were taken inside an elevator located at Buggingerstrasse, Freiburg (SWFR Dormitory), to examine changes in pressure and acceleration during vertical motion. The laptop was placed flat on the elevator floor, and the Nicla Sense Board ME was positioned sideways next to it to maintain stability. Data was recorded every 100 ms during the travel from the ground floor to the 10th floor and during the travel back to the ground floor. For both directions of travel, Z axis acceleration, pressure, and timestamps were collected. The upward run was measured at 22°C and the downward run at 20°C.

Figure 4: Setup for task 3: The Nicla Sense Board ME is positioned on the notebook with its z axis facing upward, and the notebook rests flat on the elevator floor.

4 Results and Discussion

Task 1

In the first task, the pressure sensor readings under stable conditions are analysed. The Nicla Sense ME Board was placed flat on a steady table, and the pressure(hPA) was recorded at 24°C. We recorded upto 1000 readings each and plotted it using Python(Pyplot) to analyse if there is any drift in the readings.

From Figure 5 it is observed that there is no major drift recorded in the sensor, although there is minor drift recorded it is less significant. The value under stable conditions vary upto $\pm 0.09 hPa$. This can be caused due to various reasons including environment factors and sensor calibration.

The virtual sensor `SENSOR_ID_BARO` already provides fully compensated pressure values in hPa. Therefore, no raw-to-pressure conversion was required in software.

The mean value of the observed pressure data is $978.78 hPa$. Now this is used to find the offset of each data point by the below formula:

$$Offset = Mean - ActualValue \tag{4}$$

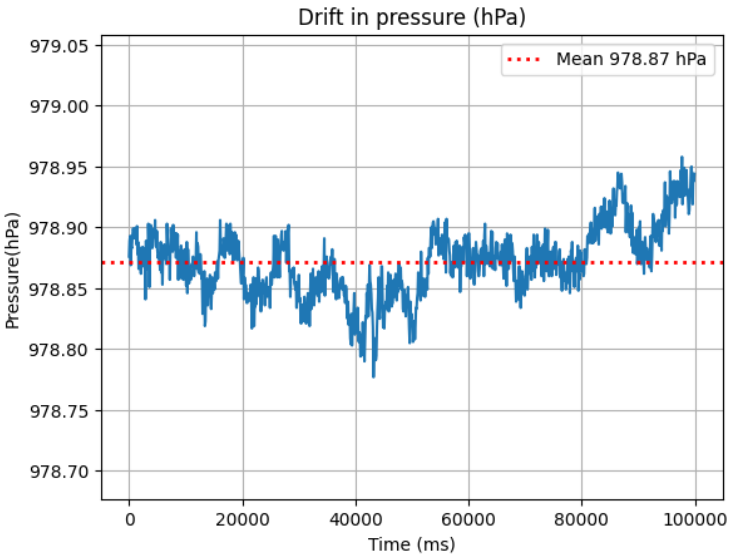

Figure 5: Pressure sensor readings at 24°C plotted to observe drift

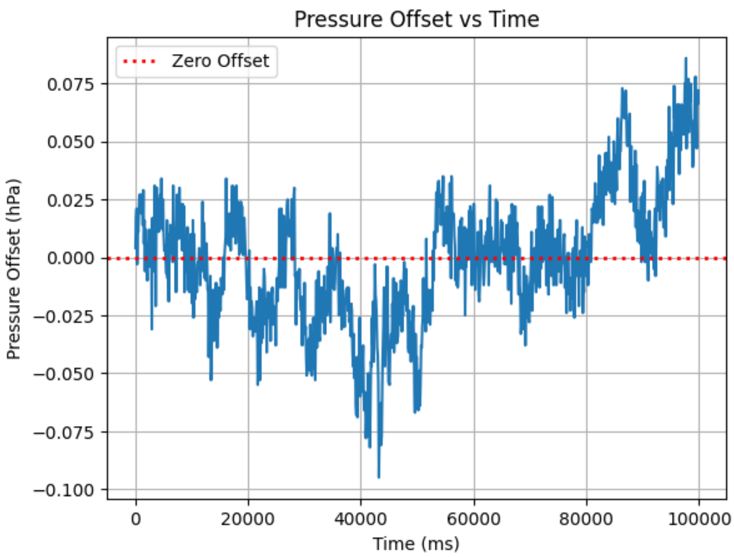

Figure 6: Pressure Offset Calculated over the 1000 readings from formula 4

From Figure 6 we see that offset readings fluctuate slightly around zero with a maximum deviation of about ± 0.09 hPa, showing small random variations rather than systematic drift. These variations are consistent with normal sensor noise and minor environmental influences and this is expected.

Table 1: Summary of the Pressure Data Observation under Stable Conditions at 24°C

Parameter	Value
Mean Pressure (hPa)	978.872
Standard Deviation of Offset (hPa)	0.022
Offset Range (hPa)	0.086 to -0.095

The histogram of the offset values is roughly symmetric centered close to zero, indicating that the noise behaves approximately Gaussian. Most values lie within about ± 0.03 hPa, with only a

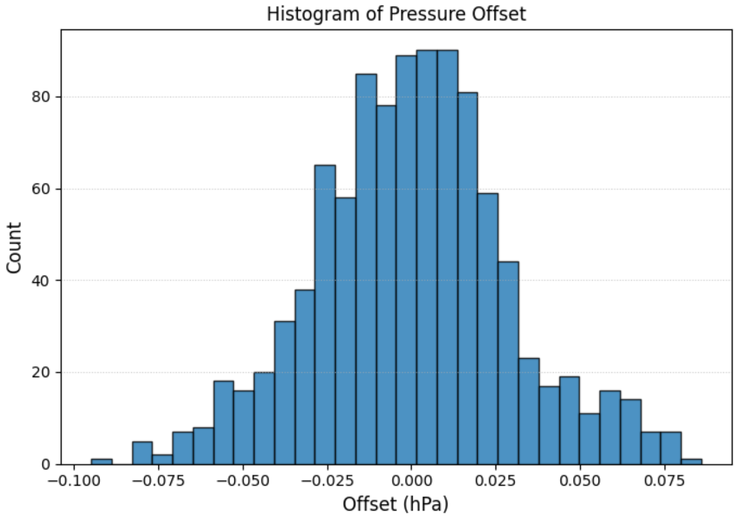

Figure 7: Histogram of the calculated offset in under stable condition pressure sensor observation

few outliers reaching ±0.08 hPa. This is expected as per the BMP390 datasheet, which specifies a relative accuracy of ±0.03 hPa, and the observed variations fall within the expected noise and environmental fluctuations.[4]. From Table 1 the pressure sensor shows a very stable output under stationary conditions, with a small standard deviation of 0.022 hPa. The offset range remains narrow at -0.095 to 0.086 hPa, which aligns well with the expected accuracy from the BMP390 datasheet [4]

Task 2

In the first task, the acceleration sensor readings under stable conditions were analyzed. The Nicla Sense ME board was placed flat on a steady table, and the acceleration (ms^{-2}) in all three directions (X, Y and Z) was recorded at 24°C. We recorded upto 1000 readings each and plotted it using Python(Pyplot) to analyse if there is any drift in the readings.

From Figure 8 it is observed that the acceleration data remains almost stable over the 1000 recorded samples, and no noticeable drift is observed. Acceleration under stable conditions is expected to be zero along the X and Y axes and proving that we observe that the X and Y axes showed only small fluctuations around their respective offset values, while the Z axis stayed close to the expected gravitational acceleration of approximately $9.8\,\mathrm{m/s^2}$. The minor variations in all axes are consistent with normal sensor noise under steady operating conditions.

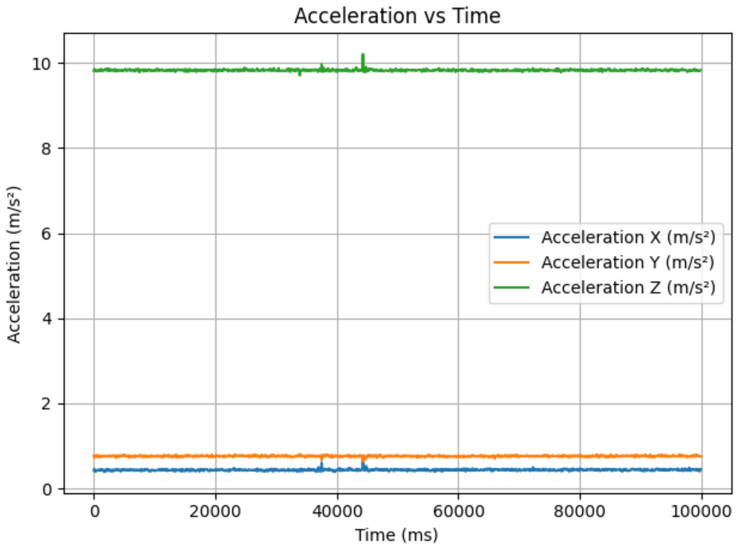

Figure 8: Acceleration values in X, Y, and Z directions recorded under stable conditions at 24°C

$$\text{Mean acceleration in X direction } (\mathrm{m/s^2}): \; 0.437$$
$$\text{Mean acceleration in Y direction } (\mathrm{m/s^2}): \; 0.761$$
$$\text{Mean acceleration in Z direction } (\mathrm{m/s^2}): \; 9.829$$

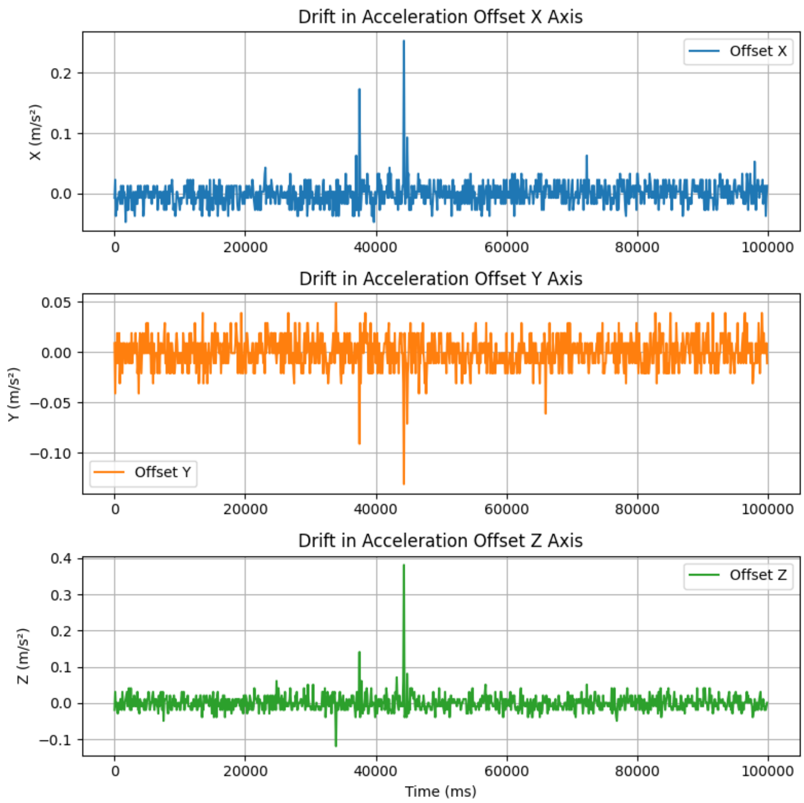

Figure 9: Drift in the acceleration offsets for the X, Y, and Z axes under stable conditions

The Offset in the acceleration sensor along the three axes is also calculated similar to the pressure sensor offset from Formula 4. From Figure 9 the acceleration offsets in the X, Y, and Z directions remain stable over time, with only small fluctuations around their respective mean values. Occasional spikes are visible in all three axes, but these appear as isolated outliers rather than systematic drift. Overall, the behavior matches the expected noise characteristics of the accelerometer under steady conditions.

Table 2: Summary of the Acceleration Sensor Data Observation under Stable Conditions at 24°C

Axis	Mean (m/s^2)	Std of Offset	Range of Offset
X	0.437	0.019	0.253 to -0.047
Y	0.761	0.016	0.049 to -0.131
Z	9.829	0.023	0.381 to -0.119

From Table 2 the measured standard deviations of 0.019 m/s^2 (X), 0.016 m/s^2 (Y), and 0.023 m/s^2 (Z) remain well within the noise levels expected for the BHI260AP accelerometer, which specifies small fluctuations around the mean under static conditions. The offset ranges observed for each axis also stay within a narrow band, indicating that the sensor performance aligns with the datasheet specifications and shows no evidence of drift.

Task 3

In the third task, the acceleration sensor readings and the pressure sensor readings are taken for the elevator ride from ground floor to tenth floor in both upwards and downwards directions. The upward ride of the elevator was done at 22°C and downward direction at 20°C. The height of the building is computed in two ways, one using the barometric formula to calculate the height with the change in pressure recorded and the other by double integrating the acceleration to obtain the height.

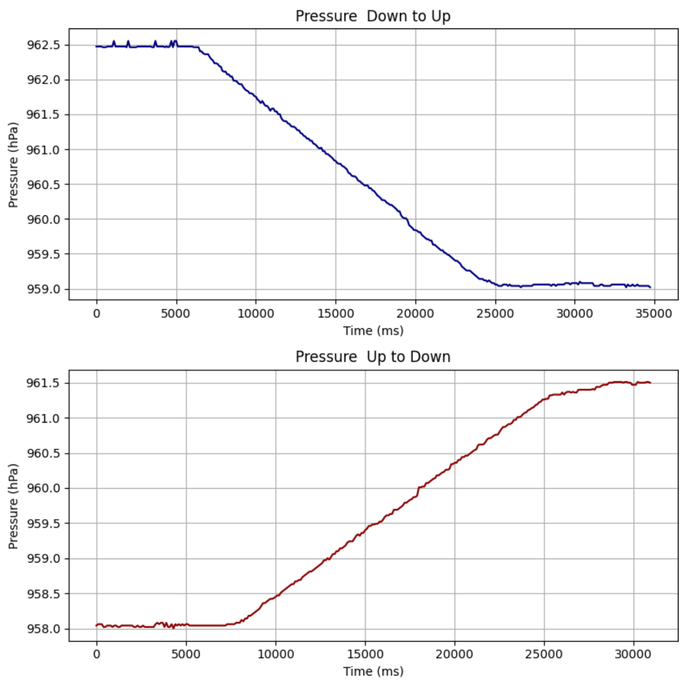

Figure 10: Pressure (hPa) variation comparison between upward and downward direction travel of the elevator

Figure 10 shows the variation of pressure for the upward and downward movement of the elevator. Using the Barometric Formula (Equation 3), the height is computed as

$$h = -\frac{RT}{gM}\ln\left(\frac{P}{P_0}\right) \tag{5}$$

and plotted using python.

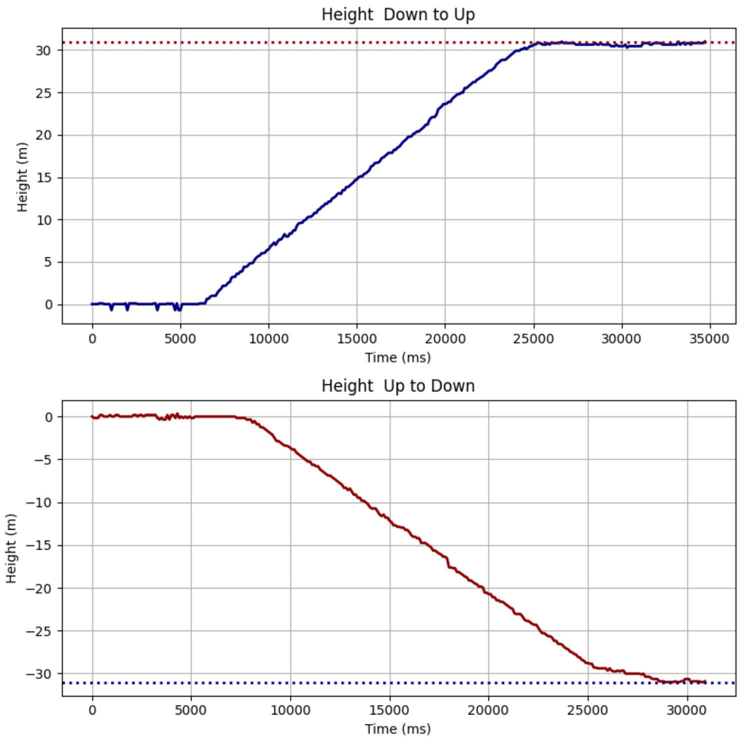

Figure 11: Height calculated using the barometric formula with the recorded pressure variation in both upward and downward directions

It is seen that the height calculated using Equation 5 for upward and downward directions result in the following:

$$\text{Height of Building (calculated using upward direction pressure data)} = 31.001 \text{ meters}$$
$$\text{Height of Building (calculated using downward direction pressure data)} = 31.006 \text{ meters}$$

$$\text{Average Height of Building by Pressure Data} = \frac{31.001 + 31.006}{2} = 31.0035 \text{ meters} \tag{6}$$

Now we attempt to calculate the height of the building by double integration the acceleration data obtained during the upward and downward movement of the elevator. The formula is as follows:

$$Distance = \int Velocity.dt = \int\int Acceleration.dt \tag{7}$$

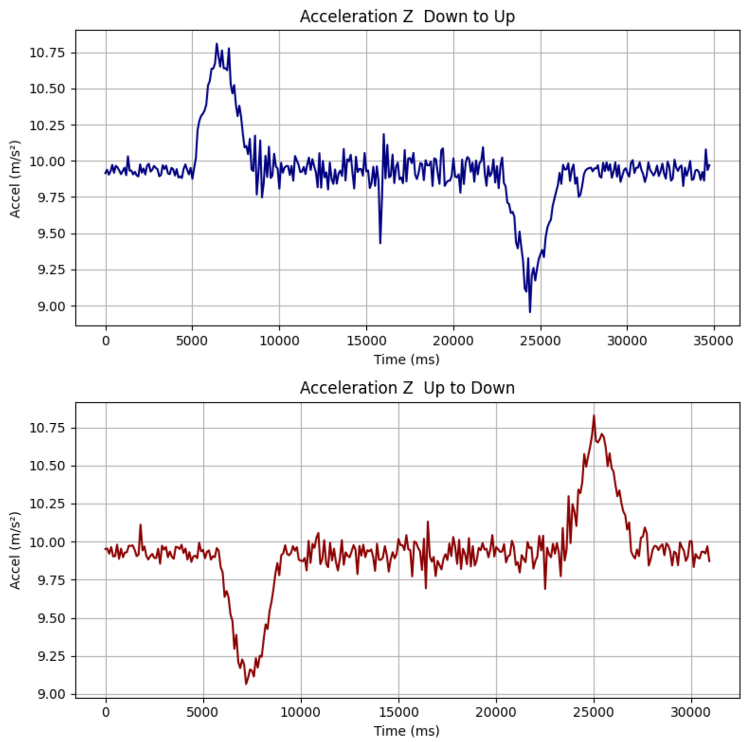

Figure 12: Acceleration data recorded in upward and downward directions of travel of the elevator

Figure 12 illustrates the variation in acceleration during the ascent and descent of the elevator. The sharp peaks correspond to the initiation and termination of elevator motion, where changes in acceleration occur. During periods of uniform motion, the acceleration remains constant because no additional forces act to change the velocity. The acceleration due to gravity is present throughout the entire motion, including when the elevator is stationary or moving at a constant velocity. This behavior is consistent for both upward and downward travel.

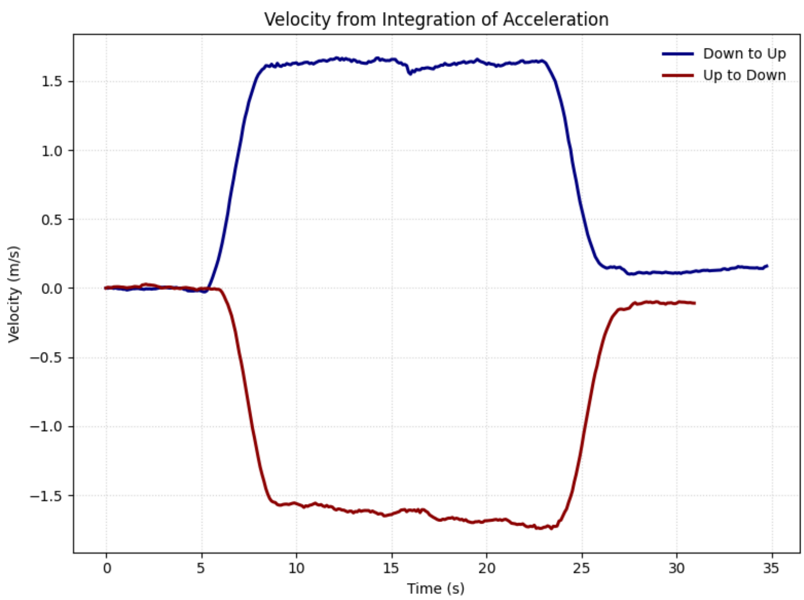

Figure 13: Velocity calculated using Equation 7

Figure 13 presents the velocity profiles obtained by integrating the measured acceleration data for both directions of elevator travel. The integration reveals the characteristic phases of motion. During the start of each movement, the velocity increases or decreases rapidly, reflecting the acceleration peaks observed earlier. Once the elevator reaches uniform motion, the velocity remains nearly constant, indicating negligible net acceleration. As the elevator approaches its destination, the velocity returns smoothly toward zero as the system decelerates. The overall patterns for upward and downward travel are similar, with opposite signs reflecting the direction of motion.

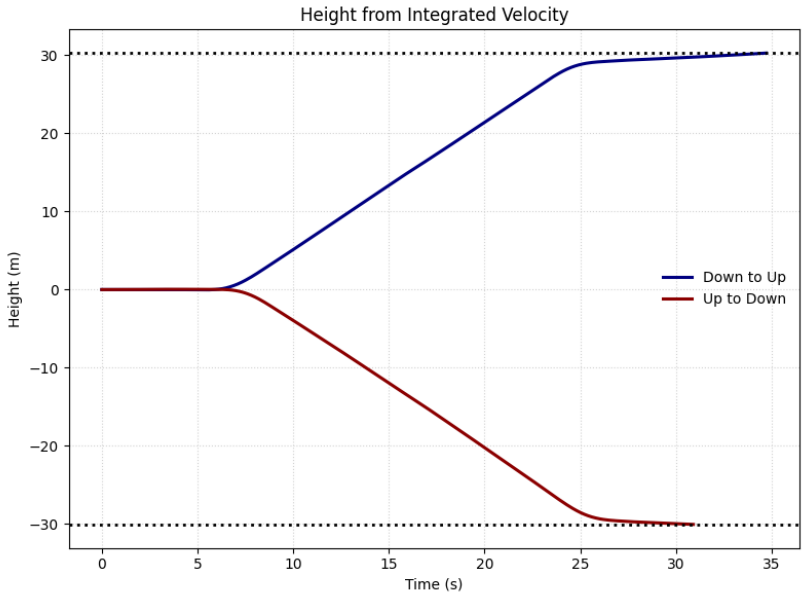

Figure 14: Height calculated using Equation 7

Figure 14 shows the height profiles obtained by integrating the velocity curves for both directions of travel. The integrated height increases steadily during the upward motion and decreases symmetrically during the downward motion, reflecting the nearly constant velocity observed during the main travel phase.

$$\text{Height of Building (calculated using upward direction acceleration data)} = 30.29 \text{ meters}$$
$$\text{Height of Building (calculated using downward acceleration direction data)} = 30.07 \text{ meters}$$

$$\text{Average Height of Building by Acceleration Data} = \frac{30.29 + 30.07}{2} = 30.184 \text{ meters} \tag{8}$$

Table 3: Summary of Building Height Calculations from Pressure and Acceleration Data

Method	**Upward (m)**	**Downward (m)**	**Average (m)**
Pressure Based Height	31.001	31.006	31.004
Acceleration Based Height	30.294	30.074	30.184

The results show that the height estimated using the pressure sensor is highly consistent between the upward and downward rides, with only a 0.005 m difference and an average of approximately 31.004 m. This agrees well with the specified relative accuracy of the BMP390 of about ± 0.03 hPa, which corresponds to a height resolution of roughly ± 0.3 m [4]. In contrast, the height obtained from double integration of the acceleration data is lower (30.184 m on average) and shows a larger deviation between directions. This behavior is expected, since even small offsets and noise in the accelerometer accumulate during numerical integration, which leads to drift in the derived velocity and position [6]. Overall, the barometric method provides the more accurate and stable estimate of the building height in this experiment.

5 Summary

The experiments demonstrated the performance of the accelerometer and pressure sensor on the Nicla Sense ME board under both stationary and dynamic conditions. The pressure sensor exhibited stable readings with minimal drift, consistent with the specified accuracy of the BMP390 [4]. The accelerometer showed low noise and predictable offsets in all three axes, matching the characteristics stated for the BHI260AP [6]. During the elevator measurements, the barometric method yielded consistent height estimates for upward and downward travel, with an average height of approximately 31.004 m. In contrast, height estimation based on double integration of acceleration showed larger deviations due to accumulated sensor noise and offset. Overall, pressure based height estimation proved more reliable for this experiment, while the acceleration data effectively captured the dynamic behavior of the elevator motion.

References

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[2] I. Schönfeld, J. Kieninger, S. J. Rupitsch, *Sensors Lab Course*. University of Freiburg. Winter term 2025/26.

[3] *Conrad - Accelerometer*, https://www.conrad.de/de/ratgeber/industrie-40/mems.html

[4] *BMP390 Datasheet: https://www.boschsensortec.com/media/boschsensortec/downloads/datasheets/bst-bmp390-ds002.pdf*

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[6] *BHI260AP Datasheet : https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst bhi260ap-ds000.pdf*